

25.1 Prediction versus intervention

A regression of Y on X answers a question about *seeing*: among units in which $X = x$ was observed, what is Y typically? The answer is $E(Y \mid X = x)$, or its best linear approximation $L(Y \mid X)$ (Theorem 6.11.1), which least squares estimates whatever produced the data (Proposition 6.11.3). A decision asks about *doing*: if X were set to x by an outside agent, what would Y typically be? This section makes the second question precise and shows, in the simplest model, that the answers differ, that more data do not close the gap, and that the better predictor can be the worse guide to action.

25.1.1 Structural causal models

A joint distribution says how variables co-vary, not which would move if another were changed. For that we need a model of the mechanism, a list of the processes that generate each variable. The following definition goes back to Wright (1921) and Haavelmo (1943).

Definition 25.1.1 — Structural causal model

Let $V_1, \dots, V_m$ be random variables. A **structural causal model** (SCM) for them consists of

- (a) for each j , a set $\text{pa}(j) \subseteq \{1, \dots, j-1\}$ of indices, the **parents** of V_j ;
- (b) for each j , a function f_j , and mutually independent random variables $U_1, \dots, U_m$, the **disturbances**, such that

$$V_j = f_j(V_{\text{pa}(j)}, U_j), \quad j = 1, \dots, m, \quad (25.1.1)$$

where $V_{\text{pa}(j)}$ is the vector of parents of V_j .

Each line of Eq. 25.1.1 is an **assignment**: it says how nature sets V_j , not merely that an equation holds. Solving the assignments in the order $j = 1, \dots, m$ writes every V_j as a function of $U_1, \dots, U_j$. For a variable $X = V_k$ and a value x , the **intervention** $\text{do}(X = x)$ replaces the k th assignment by $V_k = x$ and keeps every other assignment and every disturbance unchanged. The distribution of the variables in the modified model is the **interventional distribution**, and expectations under it are written $E(Y \mid \text{do}(X = x))$. The model is **linear** if each $f_j(\mathbf{v}, u) = c_j + \sum_{l \in \text{pa}(j)} b_{jl}v_l + u$ with constants c_j, b_{jl} , and disturbances with mean zero and finite variance.

Section 25.3 draws the parent sets as a graph. The substantive assumption is that an intervention changes one assignment and leaves the other mechanisms intact. Conditioning and intervening are then different operations: $E(Y \mid X = x)$ looks at the part of the *original* distribution where $X = x$ happened, while $E(Y \mid \text{do}(X = x))$ describes a *new* distribution in which $X = x$ was imposed.

25.1.2 A common cause

The simplest model in which the two differ has a background variable Z that influences both X and Y :

$$Z = c_Z + U_Z, \quad X = c_X + \alpha Z + U_X, \quad Y = c_Y + \theta X + \gamma Z + U_Y, \quad (25.1.2)$$

with independent disturbances of variances $\sigma_Z^2 > 0$, $\sigma_X^2 > 0$ and σ_Y^2 (Figure 25.1.1). The coefficient θ is what the assignment for Y does with its argument X .

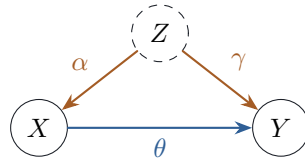


Figure 25.1.1. The model Eq. 25.1.2. An arrow points from each parent to its child and is labelled with its coefficient. The dashed circle marks a variable that is not recorded in Example (Clinic visits and health).

Proposition 25.1.1 — Seeing and doing with a common cause

In the model Eq. 25.1.2:

- (a) $E(Y \mid \text{do}(X = x)) = c_Y + \gamma c_Z + \theta x$, so the interventional slope is θ ;
- (b) the best linear predictor $L(Y \mid X)$ has slope

$$\beta^* = \theta + \frac{\gamma \alpha \sigma_Z^2}{\alpha^2 \sigma_Z^2 + \sigma_X^2}; \quad (25.1.3)$$

- (c) the coefficient of X in $L(Y \mid X, Z)$ is θ ;
- (d) if the disturbances are normal, $E(Y \mid X = x) = L(Y \mid X)$ evaluated at x , so the slopes of seeing and doing differ whenever $\gamma \alpha \neq 0$.

Proof.

- (a) Under $\text{do}(X = x)$ the assignment for Z is unchanged and does not involve X , so $Z = c_Z + U_Z$ still, and $Y = c_Y + \theta x + \gamma Z + U_Y$. Take expectations.
- (b) By Theorem 6.11.1 the slope is $\text{Cov}(X, Y) / \text{Var}(X)$. Here $\text{Var}(X) = \alpha^2 \sigma_Z^2 + \sigma_X^2$ and $\text{Cov}(X, Y) = \theta \text{Var}(X) + \gamma \text{Cov}(X, Z) = \theta \text{Var}(X) + \gamma \alpha \sigma_Z^2$.
- (c) The error $Y - c_Y - \theta X - \gamma Z = U_Y$ has mean zero and is uncorrelated with X and Z , which are functions of U_Z and U_X . So $c_Y + \theta X + \gamma Z$ satisfies the population normal equations of Theorem 6.11.1 and is the projection.
- (d) (X, Y) is a linear function of the normal vector (U_Z, U_X, U_Y) , hence bivariate normal, and its conditional mean is the best linear predictor (Theorem 3.5.2).

■

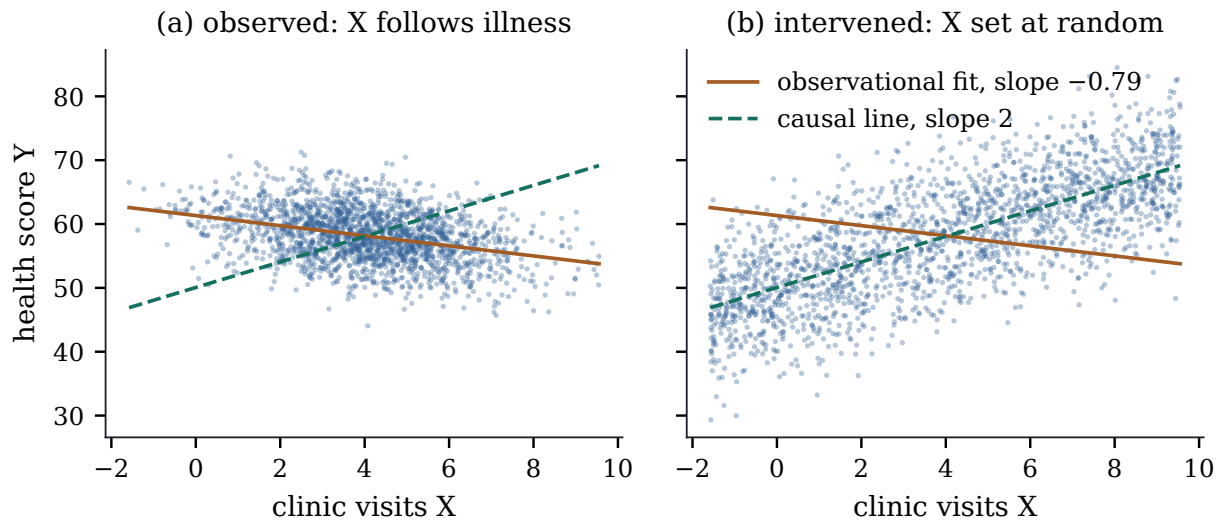


Figure 25.1.1: The model of [Example \(Clinic visits and health\)](#). (a) Observational data: the fitted line slopes down. (b) Data in which the number of visits is set at random by the analyst: the same mechanism for Y , but now X carries no information about illness. The solid line is the observational fit in both panels; the dashed line has the causal slope $\theta = 2$.

The extra term in [Eq. 25.1.3](#) is association that X borrows from Z . Part (c) is the familiar remedy, including the common cause as a regressor, and it needs Z to be recorded. [Section 25.4](#) says in general which variables must be included.

Example. (Clinic visits and health)

In a synthetic population, Z is the severity of a chronic illness, which nobody records; X is the number of clinic visits in a year; and Y is a health score measured at the end of the year. Sicker people visit more ($\alpha = 1.5$) and end the year less healthy ($\gamma = -6$), and each visit helps ($\theta = 2$). With $\sigma_Z = \sigma_X = 1$ and $\sigma_Y = 2$, [Eq. 25.1.3](#) gives

$$\beta^* = 2 + \frac{(-6)(1.5)}{1.5^2 + 1} = 2 - \frac{9}{3.25} = -0.769.$$

In a sample of $n = 2000$ people the least squares slope is -0.790 : each extra visit goes with a *lower* health score. A second data set in which visits are assigned at random, which is an intervention on X , has least squares slope 1.999 ([Figure 25.1.2](#)). A health authority that capped visits on the strength of the first regression would harm its patients.

```
import numpy as np

rng = np.random.default_rng(2501)
n = 2000
alpha, theta, gamma = 1.5, 2.0, -6.0          # Z -> X, X -> Y, Z -> Y

def draw(n, do_x=None):
    """Draw from the model; with do_x, X is set by the analyst instead of by Z."""
    Z = rng.normal(size=n)                    # illness severity, never recorded
    X = 4 + alpha * Z + rng.normal(size=n) if do_x is None else do_x
    Y = 50 + theta * X + gamma * Z + rng.normal(scale=2.0, size=n)
    return X, Y
```

```

x_obs, y_obs = draw(n)                                # observational data
x_do = rng.uniform(x_obs.min(), x_obs.max(), size=n)   # X assigned at random
x_int, y_int = draw(n, do_x=x_do)                     # interventional data

slope_obs, icpt_obs = np.polyfit(x_obs, y_obs, 1)
slope_int, icpt_int = np.polyfit(x_int, y_int, 1)
print(f"slope from observational data: {slope_obs:.3f}")
print(f"slope from interventional data: {slope_int:.3f}    (theta = {theta})")

```

[Run this code online](#)

25.1.3 A good predictor can be a bad causal model

The observational line in [Example \(Clinic visits and health\)](#) is the best linear predictor of health from visits in the population that generated the data, and the causal line, with slope θ through the same means, predicts *worse* there. On the observational data the fitted line has mean squared error 15.0 and the causal line 40.7, which is worse than predicting every person by the sample mean (the variance of Y is 17.1). On the interventional data the positions reverse: 123.3 for the fitted line and 39.2 for the causal line.

```

def mse(x, y, slope, icpt):
    return np.mean((y - icpt - slope * x) ** 2)

causal_icpt = np.mean(y_obs) - theta * np.mean(x_obs) # causal slope, fitted level
for label, x, y in [("observational", x_obs, y_obs), ("interventional", x_int, y_int)]:
    print(f"{label:15s} MSE: regression line {mse(x, y, slope_obs, icpt_obs):6.2f},"
          f" causal line {mse(x, y, theta, causal_icpt):6.2f}")

```

[Run this code online](#)

Observationally, any line through the means other than the projection loses exactly $(\text{slope} - \beta^*)^2 \text{Var}(X)$ ([Exercise \(B1\)](#)). Under intervention a large X no longer signals a large Z , so the part of the slope borrowed from Z becomes pure error: the causal line describes the mechanism both regimes share, the regression line a distribution that the intervention destroys.

Confounding by a common cause is one of three standard ways in which a predictor misleads about action. In **reverse causation** Y causes X , and X can predict Y perfectly while setting it does nothing ([Exercise \(B2\)](#)): a thermometer predicts the temperature, but moving its needle does not warm the room. In **selection** the data contain only units selected by a common effect of X and Y , which creates association without effect ([Section 25.5](#)).

More data do not help: the least squares slope converges to β^* , not θ (Proposition 6.11.3), and no procedure using only the distribution of (X, Y) can do better, because models with different θ can produce the same distribution ([Exercise \(B3\)](#)). Causal conclusions need either an intervention, as in panel (b) of [Figure 25.1.2](#), or assumptions about the mechanism that link the interventional distribution to the observational one.

25.1.4 When seeing is doing

Conditioning and intervening agree for every mechanism when nothing in the model influences X .

Proposition 25.1.2 — An exogenous cause

In a structural causal model, suppose $X = V_k$ has no parents, so that $X = f_k(U_k)$. Then a version of the conditional distribution of the other variables given $X = x$ is their distribution under $\text{do}(X = x)$; in particular, when $E|Y| < \infty$, $E(Y \mid X = x) = E(Y \mid \text{do}(X = x))$ for almost every x (with respect to the distribution of X), and for every x with $\Pr(X = x) > 0$.

Proof. Solve the assignments in order. Every variable is a function $h(X, \mathbf{U}_{-k})$ of X and the disturbances $\mathbf{U}_{-k}$ other than U_k , because U_k enters the other assignments only through X . The same function describes the model under $\text{do}(X = x)$, with x in place of X , since only the k th assignment changed. Now $X = f_k(U_k)$ is independent of $\mathbf{U}_{-k}$, and for independent X and $\mathbf{U}_{-k}$ a version of the conditional distribution of $h(X, \mathbf{U}_{-k})$ given $X = x$ is the distribution of $h(x, \mathbf{U}_{-k})$ (Billingsley 1995), which is the interventional distribution. ■

A randomized experiment makes the treatment exogenous by construction: the random device is the only parent of X and influences nothing else, so what is seen is what would be done. [Section 25.2](#) turns this into exact finite-sample statements that need no structural model.

25.1.5 Exercises

25.1.5.1 A. Check your understanding

Exercise A1.

In the model [Eq. 25.1.2](#) with $\theta = 2$, $\gamma = -6$ and $\sigma_Z = \sigma_X = 1$, for which values of α is the observational slope β^* exactly zero? What would an analyst who saw $\beta^* = 0$ be tempted to conclude?

25.1.5.2 B. Practice

Exercise B1.

Let (x_i, y_i) , $i = 1, \dots, n$, be any data with least squares slope $\hat{\beta}$. For a number b , let $\ell_b(x) = \bar{y} + b(x - \bar{x})$ be the line with slope b through the means. Show that

$$\frac{1}{n} \sum_i (y_i - \ell_b(x_i))^2 = \frac{1}{n} \sum_i (y_i - \ell_{\hat{\beta}}(x_i))^2 + (b - \hat{\beta})^2 \frac{1}{n} \sum_i (x_i - \bar{x})^2,$$

and deduce the ordering of the two observational mean squared errors in [Example \(Clinic](#)

visits and health).

Solution. Write $y_i - \ell_b(x_i) = (y_i - \ell_{\hat{\beta}}(x_i)) + (\hat{\beta} - b)(x_i - \bar{x})$. The cross term is $(\hat{\beta} - b) \sum_i \hat{e}_i(x_i - \bar{x})$, where $\hat{e}_i$ are the least squares residuals. The residuals are orthogonal to $\mathbf{1}$ and to $\mathbf{x}$ (Theorem 6.4.1), so the cross term is zero, and squaring gives the identity. With $b = \theta = 2$ the causal line through the means loses $(2 - \hat{\beta})^2$ times the variance of the visits, which is why its mean squared error exceeds that of the fitted line. The script checks the identity to rounding error.

Exercise B2.

Let $Y = c + U_Y$ and $X = \delta Y + U_X$, with independent disturbances of variances σ_Y^2 and σ_X^2 . Find the slope of $L(Y | X)$, the squared correlation of X and Y , and $E(Y | \text{do}(X = x))$. Show that the squared correlation can be arbitrarily close to one while setting X has no effect on Y .

Exercise B3.

Take $c_Z = c_X = c_Y = 0$ in Eq. 25.1.2 with normal disturbances, and suppose only (X, Y) is observed. Show that the parameter values

$$(\alpha, \theta, \gamma, \sigma_Z^2, \sigma_X^2, \sigma_Y^2) = (1, 1, 0, 1, 1, 1) \quad \text{and} \quad (1, 0, \frac{4}{3}, \frac{3}{2}, \frac{1}{2}, \frac{1}{3})$$

give the same distribution of (X, Y) , although X has an effect on Y in the first model and none in the second.

Solution. Both distributions are bivariate normal with mean zero, so it suffices to compare covariance matrices, using $\text{Var}(X) = \alpha^2 \sigma_Z^2 + \sigma_X^2$, $\text{Cov}(X, Y) = \theta \text{Var}(X) + \gamma \alpha \sigma_Z^2$ and $\text{Var}(Y) = \theta^2 \text{Var}(X) + \gamma^2 \sigma_Z^2 + 2\theta \gamma \alpha \sigma_Z^2 + \sigma_Y^2$. In the first model these are 2, 2 and $2 + 1 = 3$. In the second they are $\frac{3}{2} + \frac{1}{2} = 2$, $\frac{4}{3} \cdot \frac{3}{2} = 2$ and $\frac{16}{9} \cdot \frac{3}{2} + \frac{1}{3} = 3$. The covariance matrices agree, so no function of the (X, Y) data can distinguish $\theta = 1$ from $\theta = 0$.

25.1.5.3 C. Going deeper

Exercise C1.

Take $c_Z = c_X = c_Y = 0$ and normal disturbances in Eq. 25.1.2, and let Σ be any 2×2 positive definite matrix, with entries $v = \Sigma_{11}$, $c = \Sigma_{12}$ and $w = \Sigma_{22}$. Show that for **every** real θ there are values of $(\alpha, \gamma, \sigma_Z^2, \sigma_X^2, \sigma_Y^2)$, with $\sigma_Z^2 > 0$, $\sigma_X^2 > 0$ and $\sigma_Y^2 \geq 0$, for which $(X, Y) \sim N_2(\mathbf{0}, \Sigma)$. Hence the observational distribution constrains the causal effect not at all: Exercise (B3) exhibits two values of θ compatible with one distribution, and in fact no value is ever excluded.

Solution. Put $\sigma_Z^2 = 1$ and $\alpha = \sqrt{t}$ for a number $t \in (0, v)$ to be chosen, and $\sigma_X^2 = v - t > 0$, so that $\text{Var}(X) = \alpha^2 \sigma_Z^2 + \sigma_X^2 = v$ whatever t is. Take $\gamma = (c - \theta v)/\sqrt{t}$, so that, by the covariance formulas in the solution to Exercise (B3), $\text{Cov}(X, Y) = \theta v + \gamma \alpha \sigma_Z^2 =$

$\theta v + (c - \theta v) = c$. The same formulas leave

$$\sigma_Y^2 = w - \theta^2 v - \gamma^2 - 2\theta(c - \theta v) = w + \theta^2 v - 2\theta c - \frac{(c - \theta v)^2}{t} =: \varphi(t).$$

As $t \uparrow v$, and using $(c - \theta v)^2/v = c^2/v - 2\theta c + \theta^2 v$,

$$\varphi(t) \longrightarrow w + \theta^2 v - 2\theta c - \frac{c^2}{v} + 2\theta c - \theta^2 v = w - \frac{c^2}{v} > 0,$$

the last inequality because Σ is positive definite. So $\varphi(t) > 0$ for some $t < v$; fix such a t and set $\sigma_Y^2 = \varphi(t)$. All three variances are admissible, and (X, Y) is a linear function of a normal vector, hence normal (Theorem 3.3.1) with mean zero and the prescribed covariance matrix. Every θ is therefore compatible with every observational distribution: not only can the sign of the effect not be recovered, neither can its size.

Exercise C2.

The impossibility in [Exercise \(C1\)](#) is a fact about normal disturbances, not about observational data. Let U_X and U_Y be independent, each taking the values ± 1 with probability $\frac{1}{2}$, and let the mechanism be $X = U_X$, $Y = X + U_Y$. Show that no linear structural causal model ([Definition 25.1.1](#)) in which Y has no parents and $X = c + \delta Y + U$, with U independent of Y , reproduces the joint distribution of (X, Y) , whatever the law of U and whatever c and δ are. So here the direction of the arrow *is* visible in the data.

Solution. Under the stated mechanism (X, Y) takes the four values $(1, 2)$, $(1, 0)$, $(-1, 0)$, $(-1, -2)$, each with probability $\frac{1}{4}$; in particular $Y \in \{-2, 0, 2\}$ with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. Suppose $X = c + \delta Y + U$ with U independent of Y . Then the conditional law of $X - c - \delta Y$ given $Y = y$ is the law of U for every y with $\Pr(Y = y) > 0$. But given $Y = 2$ the only possible value of X is 1, so that conditional law is degenerate at $1 - c - 2\delta$; while given $Y = 0$, X equals 1 or -1 with probability $\frac{1}{2}$ each, so the conditional law puts mass $\frac{1}{2}$ on each of $1 - c$ and $-1 - c$ and is not degenerate. A degenerate law and a two-point law cannot both be the law of U , a contradiction.

The contrast with [Exercise \(C1\)](#) is the whole point. Two independent non-normal disturbances leave a trace in the joint law that tells a cause from an effect; two independent normal ones do not, because a normal vector is determined by its first two moments and those are symmetric in the two directions. Methods that exploit non-normality to orient arrows rest on this observation.

Exercise C3.

In [Eq. 25.1.2](#), suppose an outside agent sets X to a random value $\tilde{X}$ with mean m and variance $\nu > 0$, independent of (U_Z, U_X, U_Y) ; this is $\text{do}(X = \tilde{X})$, and the observational regime is *not* of this form.

- (a) Show that in the new regime the line $\ell(x) = a + bx$ has mean squared prediction error

$$\mathbb{E}\{Y - \ell(\tilde{X})\}^2 = \{k + (\theta - b)m\}^2 + (\theta - b)^2\nu + \gamma^2\sigma_Z^2 + \sigma_Y^2, \quad k = c_Y + \gamma c_Z - a.$$

- (b) Deduce that the minimum is attained exactly at $b = \theta$ and $a = c_Y + \gamma c_Z$, whatever

the agent's device is, and that the minimum value $\gamma^2\sigma_Z^2 + \sigma_Y^2$ does not depend on the device either.

- (c) Deduce that the observational line of Eq. 25.1.3 has excess error at least $(\beta^* - \theta)^2\nu$, which grows without bound as the agent spreads X more widely, and that a line that is optimal under two devices with different means is the causal line.

The causal line is thus the only line that predicts well under every intervention; the regression line is optimal for one regime, the one the intervention destroys.

Solution.

- (a) Under $\text{do}(X = \tilde{X})$ the assignments for Z and Y are unchanged, so $Z = c_Z + U_Z$ and $Y = c_Y + \theta\tilde{X} + \gamma Z + U_Y$. Hence

$$Y - \ell(\tilde{X}) = k + (\theta - b)\tilde{X} + \gamma U_Z + U_Y,$$

with $k = c_Y + \gamma c_Z - a$. The last two terms have mean zero and are independent of $\tilde{X}$, so the mean squared error is $E\{k + (\theta - b)\tilde{X}\}^2 + \gamma^2\sigma_Z^2 + \sigma_Y^2$, and $E\{k + (\theta - b)\tilde{X}\}^2 = \{k + (\theta - b)m\}^2 + (\theta - b)^2\nu$.

- (b) Both squared terms are nonnegative and $\nu > 0$, so the second forces $b = \theta$ and the first then forces $k = 0$, that is $a = c_Y + \gamma c_Z$. Neither the minimizer nor the minimum involves m or ν .
- (c) With $b = \beta^*$ the second term alone is $(\beta^* - \theta)^2\nu$, and by Eq. 25.1.3 this is nonzero whenever $\gamma\alpha \neq 0$; it grows linearly in ν . If a line is optimal for devices with means $m_1 \neq m_2$ and positive variances, (b) applied to each gives $b = \theta$ and $a = c_Y + \gamma c_Z$ — the same line. The observational regime escapes (b) only because there X is *not* independent of U_Z : the association it borrows from Z is real, and worth using, exactly as long as nobody intervenes.